

Kelvin waves can raise water levels along the US West Coast

From: PBS NewsHour Mined: 2026-10-03 3 min read

Some ocean waves never curl over and crash onto a beach. Kelvin waves are broad **bulges** of water that move through the ocean and can raise the sea level along a coast.

In an AP report, experts **forecast** a series of these waves along the United States' West Coast from October through February. They predict water levels could rise by six inches to a foot as the waves travel north along California toward Oregon and Washington.

These waves are not tsunamis. They are so broad that people cannot see them as an ordinary wave, and they do not make boats bob up and down, says scientist Dillon Amaya.

The process begins with winds near the **equator**, the imaginary line around Earth's middle. Winds usually push Pacific water west toward Asia. When those winds weaken or **reverse**, a bulge of water can move east. After reaching South America, it travels along the coast to the north and south.

Kelvin waves are common during El Niño, a natural climate pattern involving warmer ocean water. Experts say the especially strong El Niño described in the report could increase their effects.

A higher sea level gives tides and storms a higher starting point. That can increase flooding and **erosion**, the wearing away of beaches. Some coastal communities already face these problems.

The waves also warm the ocean and push cold, **nutrient-rich** water deeper, affecting marine life. Their effects will depend partly on how they combine with tides, storms and other waves.

Word Treasure

bulges -- Parts that swell outward or rise above their surroundings.

forecast -- Predict what may happen.

equator -- The imaginary line around Earth's middle.

reverse -- Change to the opposite direction.

erosion -- The wearing away of material, such as beach sand.

nutrient-rich -- Containing many substances that living things need to grow.

Background you need

El Niño is a natural climate pattern involving warmer ocean water.

Erosion means the wearing away of material, including beach sand.

Break it down

Introduces Kelvin waves as broad bulges of water.

Presents the report's West Coast forecast.

Explains their appearance and how changing winds form them.

Connects the waves to El Niño and coastal effects.

Describes changes to water below the surface.

Why it matters

Understanding Kelvin waves helps explain why tides and storms can bring more water onto beaches when the sea level is already higher.

Test what you remember

1. What are Kelvin waves?

- ☐ (A) Broad bulges of ocean water
- ☐ (B) Strong gusts of coastal wind
- ☐ (C) Narrow streams of river water
- ☐ (D) Thick layers of floating ice

2. How much could water levels rise in the report's forecast?

- ☐ (A) One inch to three inches
- ☐ (B) Two feet to three feet
- ☐ (C) Six inches to a foot
- ☐ (D) Four inches to five inches

3. Which way are the waves forecast to travel along California?

- ☐ (A) West toward Asian shores
- ☐ (B) South toward South America
- ☐ (C) East toward inland mountain ranges
- ☐ (D) North toward Oregon and Washington

4. What can allow a bulge of Pacific water to move east?

- ☐ (A) The winds weakening or reversing
- ☐ (B) The coastal beaches growing wider
- ☐ (C) The ocean water becoming colder
- ☐ (D) The boats moving up and down

5. Why can a higher sea level increase flooding?

- ☐ (A) It makes beaches grow more sand.
- ☐ (B) It gives storms a higher starting point.
- ☐ (C) It makes winds blow toward Asia.
- ☐ (D) It gives boats a faster travel speed.

6. What happens to cold, nutrient-rich water as these waves pass?

- ☐ (A) It gets pushed deeper down.
- ☐ (B) It turns into floating ice.
- ☐ (C) It moves onto sandy beaches.
- ☐ (D) It gathers inside river mouths.

Answer key (try the quiz first!): 1: A 2: C 3: D 4: A 5: B 6: A

1. Dillon Amaya: He says Kelvin waves cannot be seen as ordinary waves and do not make boats bob up and down.